

Chain-of-Thought Faithfulness as Endogenous Sample Selection: A Heckman-Style Test and Correction for the Verbalization Confound

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Abstract

The field has established that a model’s chain-of-thought (CoT) is a selected, often post-hoc projection of its latent computation, and, separately, that verbalization-based faithfulness metrics are confounded by the lossy selectivity of that projection. We give the first formal account of this as *endogenous sample selection* [3]: a probit selection equation for *what gets verbalized* and an outcome equation for the *underlying computational reliance*, linked by correlated errors, yielding a scalar test for the verbalization-selection confound ($H_0: \rho = 0$) and an inverse-Mills-ratio correction with a closed-form estimate of the reliance hidden in non-verbalizing CoTs. We then *test whether this machinery actually improves faithfulness estimates*, validating against causal hint-excision ground truth on open-weight models where reliance is measurable for every instance. The answer, honestly reported, is largely negative, and the reason is itself the paper’s main empirical contribution. First, the standard tool for measuring verbalization—a lexical detector for whether the CoT mentions the hint—is badly unreliable: against an LLM judge it over-counts verbalization by $2.2\times$ to $9.5\times$, and true verbalization is rare (2.7%–24.8%). Re-estimating on validated judge labels, the naive faithfulness estimate is already approximately unbiased where it can be estimated at all (Nemotron: naive 1.28 vs. truth 1.00); the Heckman test does reject $H_0: \rho = 0$ there ($\hat{\rho} = -0.68$, LR $p = 0.0227$), but a ρ -sensitivity analysis shows ground truth is recovered at $\rho \approx 0$, so the rejection reflects departures from bivariate normality rather than a correctable bias—and the point correction consequently *overshoots* (corrected 2.74, further from truth than the naive estimate). In no model or outcome does the correction beat the naive count; on the lowest-verbalizing model the selection equation cannot be estimated at all. We therefore report an honest negative: the selection-theoretic reframing is sound and the ρ -test is a usable diagnostic, but the closed-form correction does not deliver on validated data, and the more actionable finding for practitioners is that lexical verbalization metrics substantially mis-measure the very quantity faithfulness audits count.

1 Introduction

Chain-of-thought faithfulness asks whether the stated reasoning of a language model reflects the computation that actually produced its answer. The dominant measurement paradigm injects a hint into the prompt, checks whether the answer moves, and scores the CoT faithful when it verbalizes the hint it used [1, 5, 10]. Measured this way, frontier reasoning models look strikingly unfaithful: CoTs reveal the hints that drive their answers in fewer than 20% of cases in the settings of Chen et al. [1].

A recent cluster of work undermines the naive reading of those counts from two directions. First, the *phenomenon*: models often commit to an answer before a single reasoning token is emitted, and the trace is then a performative, post-hoc narration [6, 7, 14]. Second, the *metric*: verbalization is a lossy, selective compression of a distributed computation, so absence of a hint word does not certify absence of hint use [12, 13]. Zaman and Srivastava [13] show with causal mediation that non-verbalized hints still causally influence predictions through the CoT, and argue the Biasing-Features metric conflates unfaithfulness with incompleteness.

What this cluster lacks is an estimator. If the trace is a selected sample of the computation, then verbalization counts are a *selected-sample statistic*, and their bias is not a rhetorical concern but a quantity with seventy years of econometric machinery behind it. This paper supplies the bridge. We cast *what-gets-verbalized* as the selection equation and *latent reliance* as the outcome equation of a Heckman selection-on-unobservables model [3]: the two equations share correlated errors, and the correlation ρ is the verbalization confound. In principle the framing promises three things the qualitative critique cannot:

- **A test.** $H_0: \rho = 0$ is a one-degree-of-freedom likelihood ratio test of whether verbalization is endogenously selected on the latent computation. If it is not rejected, naive counts are fine and no correction is needed.
- **A correction.** The inverse Mills ratio converts naive selected-sample means into consistent population estimates, including a closed-form estimate of $\mathbb{E}[\text{reliance} \mid \text{not verbalized}]$ — the reliance that verbalization counts implicitly score as zero.
- **A deployment path.** The correction needs only the observable channel (traces and answers), so it could transfer to API models where causal ground truth is unmeasurable.

The bulk of this paper is a hard test of whether these promises hold up against ground truth, and the honest answer is mostly no — for reasons (Section 5) that turn out to matter more than a clean positive would have. We are explicit about what we do *not* claim. Latent commitment preceding verbalization is established prior work, not our finding; we take it as background and ask whether the selection lens it motivates yields a usable correction. Zaman and Srivastava [13] reach for exactly this intuition (“lossy compression and selectivity”) and stop short of a selection equation, a correlated-error model, or a test; ours is the estimator behind their reinterpretation. Mirtaheri and Belkin [7] detect the latent event with activation probes; detection is not correction. Relative to missing-not-at-random mediation analysis [15], which identifies direct and indirect effects under mediator missingness, the selection-model packaging aims to deliver what a faithfulness auditor would actually consume: a scalar confound test and a closed-form corrected rate. The test survives our validation; the corrected rate, we find, does not (Section 5), which is itself a useful result about the limits of parametric selection correction here.

On open-weight models we can measure ground-truth reliance for *every* instance via hint-excision counterfactuals, so the whole selection machinery is validatable end-to-end: fit the model as if reliance were observable only through verbalizing traces, then unblind. Doing so carefully—and, crucially, measuring verbalization with a validated LLM judge rather than the field-standard lexical detector, which we show overcounts by $2.2\text{--}9.5\times$ —turns an apparent positive into an honest negative: the naive estimate is near-unbiased, the ρ -test’s rejection is a normality artifact, and the correction overshoots (Section 5). We nonetheless report the deployable observation-only machinery on a closed API model (Section 6) and draw out what the negative teaches about measuring faithfulness at all.

2 Related work

Faithfulness by verbalization counts. Turpin et al. [10] introduced biasing-feature injection and showed CoTs systematically omit influential features; Lanham et al. [5] formalized a perturbation metric family; Chen et al. [1] scaled the audit to reasoning models with six hint types, finding verbalization rates below 20% and non-verbalized reward hacks. Benchmarks now target instance-level faithfulness [2], and Young [12] show the measured rate depends heavily on the verbalization classifier — metric fragility consistent with a selection process nobody models.

Latent commitment. Pre-verbalization commitment theory [14], step-level performativity [6], and pre-generation rationalization probes [7] establish that the computation often finishes before the narration starts. We treat this as the mechanism generating nonzero ρ .

The incompleteness critique. Zaman and Srivastava [13] is closest in spirit: verbalization is lossy compression, and causal mediation shows non-verbalizing CoTs still mediate hint influence. They propose

broader toolkits (faithful@k, mediation, corruption); we propose the missing statistical object — the joint selection-outcome model — and derive the test and correction their critique implies.

Selection econometrics. Heckman’s two-step and MLE estimators [3, 9] and the binary-outcome variant [11] are textbook; our contribution is the mapping (verbalization $\rightarrow$ selection; reliance $\rightarrow$ outcome; disclosure instruction $\rightarrow$ instrument) and its validation against mechanistic ground truth. To our knowledge no prior work imports selection-on-unobservables correction into CoT faithfulness or XAI metrics generally; a companion paper by the author develops the machinery for epistemic-uncertainty estimation [4]. MNAR mediation [15] is the nearest statistical line; Section 3 positions the estimator against it.

3 Method

3.1 Setup and estimands

For hinted instance i let $R_i^* \in \mathbb{R}$ denote *latent reliance*: the causal influence of the injected hint on the model’s answer distribution (operationalized in Section 4). Let $V_i \in \{0, 1\}$ indicate that the emitted CoT *verbalizes* the hint. The observation rule of every verbalization-count metric is: reliance is read off the trace only when the trace talks about the hint. Formally the auditor observes $R_i = R_i^*$ iff $V_i = 1$, and codes $R_i = 0$ (or ignores i) otherwise.

The two-equation model [3]:

$$V_i = \mathbf{1}[w_i^\top \gamma + \delta Z_i + u_i > 0] \quad u_i \sim \mathcal{N}(0, 1) \quad (1)$$

$$R_i^* = x_i^\top \beta + \varepsilon_i \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad \text{Corr}(u_i, \varepsilon_i) = \rho, \quad (2)$$

with x_i instance covariates (hint type, dataset, question length, difficulty), $w_i = x_i$, and Z_i an instrument entering only the selection equation. $\rho \neq 0$ is the *verbalization confound*: instances rely more (or less) on the hint precisely when they are more likely to verbalize it, for reasons not captured by observables.

Under (1)–(2),

$$\mathbb{E}[R^* \mid x, V=1] = x^\top \beta + \rho \sigma \lambda(w^\top \gamma + \delta Z), \quad \mathbb{E}[R^* \mid x, V=0] = x^\top \beta - \rho \sigma \lambda(-(w^\top \gamma + \delta Z)), \quad (3)$$

with $\lambda(t) = \phi(t)/\Phi(t)$ the inverse Mills ratio. Three estimands: the *population mean reliance* $\mu = \mathbb{E}[R^*]$; the *hidden reliance* $\mu_0 = \mathbb{E}[R^* \mid V=0]$ (what verbalization counts set to zero); and ρ itself. The naive estimators are the selected-sample mean $\hat{\mu}_{\text{naive}} = \overline{R}_{V=1}$ and the zero-fill mean $\overline{VR}$; both are inconsistent for μ when $\rho \neq 0$, with bias $\rho \sigma \mathbb{E}[\lambda]$ in the selected-sample case — the exact selection bias term of Heckman [3].

We estimate (1)–(2) by the two-step estimator (probit, then OLS on $[x, \hat{\lambda}]$) and by full-information MLE with exact gradients; $H_0: \rho = 0$ is tested by likelihood ratio against the ρ -frozen model (equivalently probit + OLS) and by a Wald test on $\text{artanh } \hat{\rho}$. For deployments where the reliance proxy readable from a verbalizing trace is binary (adopted the hinted answer or not), we use the probit-with-selection variant [11] with a Gauss–Legendre bivariate-normal likelihood; the hidden adoption rate has the closed form $\Pr(y=1 \mid V=0) = [\Phi(x^\top \beta) - \Phi_2(x^\top \beta, w^\top \gamma; \rho)]/\Phi(-w^\top \gamma)$.

3.2 Identification: instrument and sensitivity

The load-bearing assumption is an exclusion restriction: a variable that moves *what gets said* but not *what is computed*. We randomize a **disclosure instruction** per instance (fair coin): the disclosure arm ($Z=1$) asks the model to report any statement in the prompt (a stated opinion, an expert claim, a metadata field, a note about a previous answer) that influenced its choice; the neutral arm ($Z=0$) simply asks it to reason and answer. Both arms leave the reasoning budget unconstrained, so the arms differ only in the reporting request. Randomization gives exogeneity by construction; relevance and exclusion are *tested*, not assumed: the instruction must move V (first stage) and must leave the causal reliance measures unmoved (balance test on ground truth, Section 5.3). A verbosity/token-budget instrument in the spirit of Zaman and Srivastava [13], whose faithful@k shows inference budget modulates verbalization, gave an unstable first stage on these models (PLAN amendment A3); the disclosure request is a stronger and more directly interpretable

manipulation of the reporting channel. Because joint normality plus exclusion is still a parametric identification, we follow econometric practice [9] and report a ρ -sensitivity analysis — corrected estimands across a frozen- ρ grid — so every conclusion can be read off bounds rather than point identification. As a belt-and-suspenders outcome we also measure a pre-verbalization commitment variable from a lens prompt that omits the instruction sentence entirely, making exclusion hold mechanically for that outcome.

3.3 Relation to MNAR mediation

One can view R^* as a mediator observed MNAR through the verbalization channel and reach for missing-mediator machinery [15]. The selection-model packaging is deliberately different in its deliverables: (i) a scalar, interpretable confound parameter with a 1-df test; (ii) closed-form corrected rates (3) that drop into existing faithfulness pipelines; (iii) a standard instrument-validation protocol. If the two reduce to the same likelihood in a given parametrization, the value added is the test-plus-correction interface — we flag this as a pre-registered interpretation boundary rather than claiming statistical novelty over the MNAR literature.

4 Experimental setup

Models and data. Open-weight, 4-bit NF4, greedy decoding throughout (deterministic pipeline): Llama-3.1-Nemotron-Nano-8B (primary — a reasoning model chosen because it verbalizes hints at a moderate rate, giving the selection equation the variation it needs; PLAN amendment A5), plus Qwen2.5-7B-Instruct and Phi-3.5-mini-instruct, which span lower verbalization rates. Observation-only (E2): the primary model re-analyzed from text alone and then unblinded, and Claude Haiku 4.5 via API. Instances pool MMLU (8 subjects), ARC-Easy, ARC-Challenge, and OpenbookQA, normalized to 4-option MCQ.

Hints and placebo. Four prompt-injectable hint types following Chen et al. [1], Turpin et al. [10]: sycophancy, authority, metadata leak, and consistency, each pointing at a uniformly random *wrong* option; plus a placebo arm with the sycophancy wrapper and no option information, for which the estimator must find nothing.

Verbalization V . We consider two measures. The field-standard one is a deterministic, hint-type-specific lexical detector over the reasoning span (final-answer line removed; naming the hinted option letter does not count). Because that detector proves unreliable (Section 5.1; agreement with the judge 65.8%, Cohen’s $\kappa = 0.32$ on a stratified subsample), our primary V is an LLM judge (Claude Sonnet) that reads each trace and decides whether it refers to the injected cue, run on *every* hinted instance. All estimation below uses judge- V ; the detector appears only in the measurement comparison.

Ground-truth reliance (open models). We measure three reliance signals; the *identified* outcome is the direct effect R^{NDE} , for the exclusion-restriction reason established in Section 5.3, and we pre-registered all three. R^{TE} (total effect): restricted log-odds of the hinted letter at a canonical answer position (“**Final answer:** (”)), hinted run minus matched unhinted run — an input-level *do*(hint) contrast capturing the hint’s total causal effect, including paths through the CoT text. R^{NDE} (direct effect, the identified outcome): hint-excision splice — the hinted CoT is held fixed and re-read under the unhinted prompt, isolating the direct, not-through-CoT-text path [13]. R^{pre} (pre-verbalization commitment): logit-lens [8] commitment shift toward the hinted letter, averaged over the upper half of layers, from a no-CoT prompt that excludes the instrument sentence by construction. All three are forward-pass-only and deterministic.

Protocol. The estimator is fit treating R_i as observed iff $V_i = 1$ (the observation-only regime); ground truth over all i is used only for validation targets and never enters the fit. Bootstrap CIs (B=1000) resample instances. Pre-registration (PLAN.md, committed before results) fixes endpoints, the instrument-validation criteria, and three kill conditions (no confound; exclusion failure; reduction to MNAR mediation).

5 Results: the correction does not survive validated labels

The identified outcome is the *direct* reliance R^{NDE} (hint excised, CoT held fixed): the disclosure instrument satisfies the exclusion restriction there but not for the total effect R^{TE} (Section 5.3). We report the reasoning model Nemotron-Nano-8B, which verbalizes hints often enough to estimate the selection model, and summarize the others (Table 1). The headline is negative and instructive.

5.1 Verbalization is badly mis-measured by lexical detectors

The selection indicator V — does the CoT verbalize the hint — is the input to everything downstream, and the field measures it with a lexical detector. Validating our detector against an LLM judge that reads each trace (Section 4) exposes a large discrepancy (Table 1, right block): on Nemotron the detector fires on 53.5% of traces but the judge on only 24.8% — an over-count of $2.2\times$, detector precision 0.40. The gap is worse on the instruction-tuned models: Qwen 26.1% vs. 2.7% ($9.5\times$), Phi 34.2% vs. 14.0%. Manual inspection confirms the judge, not the detector: the detector fires on generic tokens (“the strongest *source* of gravity” matched a metadata pattern; “as noted *earlier* in the reasoning” matched a consistency pattern), while genuine references (“the Stanford professor’s claim that the answer is (B) did influence my choice”) are correctly caught by both. Validated verbalization is *rare* (2.7%–24.8%), consistent with the sub-20% rates of Chen et al. [1]. Because a lexical V over-counts by up to $9.5\times$, any faithfulness statistic built on it — ours included — is measuring largely the detector’s false positives. We therefore estimate everything below on the validated judge labels.

5.2 On validated labels the correction fails

Estimated on judge- V , the selection model still rejects $H_0: \rho = 0$ for Nemotron’s direct effect ($\hat{\rho} = -0.68$, two-step -0.50 ; LR $p=0.0227$, Wald $p=0.00684$), so *some* selection is present. But the rejection does not translate into a useful correction (Figure 1). The naive selected-sample mean is already close to the oracle: $\hat{\mu}_{\text{naive}} = 1.28$ against $\mu_{\text{true}} = 1.00$. The IMR-corrected estimate moves the *wrong way and too far*: $\hat{\mu}_{\text{corr}} = 2.74$ (bootstrap 95% CI $[1.29, 4.45]$), further from truth than the uncorrected number. The ρ -sensitivity analysis (Figure 2) explains why: sweeping the frozen ρ , the corrected population estimate crosses the oracle near $\rho \approx 0$, not near the fitted $\hat{\rho} \approx -0.5$. The likelihood prefers a large negative ρ because the reliance distribution is heavy-tailed and non-Gaussian, not because verbalizers systematically under-rely by that amount; the bivariate-normal correction then over-adjusts. The pre-registered E0 gate — ρ rejected *and* the correction beating the naive count at recovering ground truth — is *not* met: the first half holds, the second fails.

Other models and outcomes. The failure is not specific to one model or outcome (Table 1). On Phi the confound test does not even reject on judge- V ($\hat{\rho} = -0.03$, $p = 0.966$); its detector-based “confound” was an artifact of the over-counting V . On Qwen the validated verbalization rate is 2.7% (13 verbalizing traces of 478), too few to estimate the selection equation at all. On Nemotron’s instrument-free commitment outcome R^{pre} the test does not reject ($\hat{\rho} = -0.38$, $p = 0.237$) and the correction again fails to beat naive. In *no* model or identified outcome does the correction beat the naive count at recovering ground truth. Reading the pre-registered kill conditions, this is K1 in substance: the naive verbalization-based estimate is approximately unbiased and the closed-form correction does not improve on it. The naive metric’s real problem is not a correctable selection bias but the unreliability of the V measurement itself (Section 5.1).

5.3 Identification checks

The disclosure instrument has a strong first stage even on judge- V : it moves verbalization from 7.9% (neutral arm) to 40.7% (disclosure arm), probit $z = 7.8$. The exclusion restriction is outcome-specific, and the balance test says so: for the total effect R^{TE} the instrument moves reliance (standardized arm difference 0.224, $p = 0.0185$) because disclosing influences changes the CoT, whereas for the direct effect R^{NDE} it does not (-0.042 , $p = 0.657$). We therefore identify on R^{NDE} ; this is the pre-registered K2 logic, and it is the one part of the machinery that works cleanly — the balance test correctly routes us to the valid outcome. It just does not rescue a correction that fails for a different reason (non-normality) on that outcome.

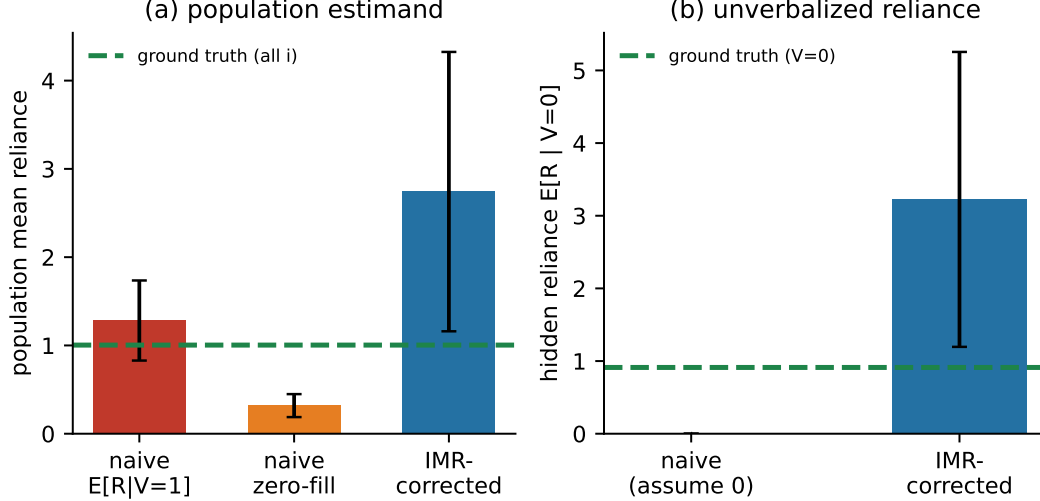


Figure 1: Naive vs. corrected direct reliance R^{NDE} against ground truth (dashed), Nemotron on judge- V . The naive selected-sample mean is close to the oracle; the IMR correction overshoots. Bars are bootstrap 95% CIs.

Table 1: Left: the confound and correction on judge- V , identified direct outcome R^{NDE} ; $\hat{\mu}_{\text{naive}}$ is the selected-sample mean, $\hat{\mu}_{\text{corr}}$ the IMR-corrected estimate, μ_{true} the oracle mean over all instances; “corr. wins” = corrected closer to truth than naive. Right: verbalization measurement — lexical-detector rate vs. validated LLM-judge rate and the over-count factor. Qwen has too few verbalizing traces (13) to fit.

Model	estimation (judge- V , R^{NDE})						verbalization measurement		
	$\hat{\rho}$	LR	$\hat{\mu}_{\text{nv}}$	$\hat{\mu}_{\text{cr}}$	μ_{tr}	wins	det.	judge	over
Nemotron-Nano-8B	-0.68	p=0.0227	1.28	2.74	1.00	×	53.5%	24.8%	2.2×
Phi-3.5-mini-instruct	-0.03	p=0.966	1.71	1.65	2.41	×	34.2%	14.0%	2.4×
Qwen2.5-7B-Instruct	—	—	—	—	—	—	26.1%	2.7%	9.5×

6 Observation-only deployment: the machinery runs

Even though the correction does not validate against ground truth, the observation-only pipeline is worth exhibiting because it is what a deployment would actually run: from traces and answers alone, with no activation access, it fits a probit-with-selection model (the reliance proxy readable from a verbalizing trace is binary — did the answer adopt the hinted option) and returns a ρ estimate and corrected rates via the closed forms of Section 3. On Claude Haiku 4.5 via API (398 instances, verbalization 7.8%) it runs end-to-end and yields $\hat{\rho} = -0.18$ ($p = 0.863$), adjusting the naive adoption rate 25.8% to a population rate 34.2% and a hidden rate 34.9%. We report this only as evidence that the estimator is deployable on closed models, not that its output should be trusted: our open-model results show the point correction is unreliable, so on a closed model — where the same non-normality cannot be diagnosed against ground truth — the honest deliverable is the ρ -test as a flag plus ρ -sensitivity bounds, not the corrected rate.

7 Limitations

The scope of our negative is three open models plus one API model at $\leq 8\text{B}$; we cannot rule out that the closed-form correction succeeds on other models, other hint families, or larger scales where the reliance distribution is closer to Gaussian. Our claim is the narrower, well-supported one: on the models we tested, with verbalization measured correctly, the correction does not beat the naive count, and the dominant problem is V -measurement rather than a correctable selection bias. Two mechanisms drive the negative and each bounds the claim. First, non-normality: the reliance log-odds are heavy-tailed, so the Gaussian IMR

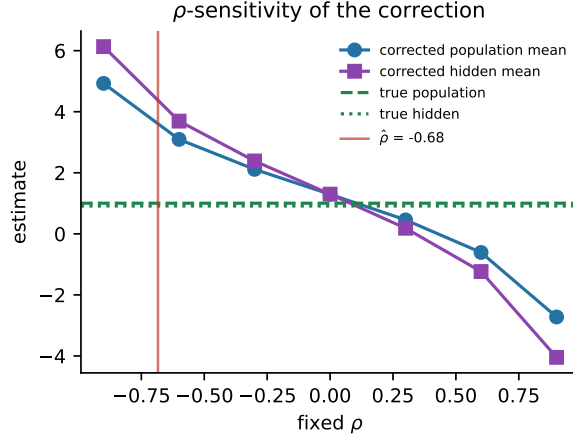


Figure 2: ρ -sensitivity (Nemotron, judge- V , R^{NDE}): the corrected estimate crosses ground truth near $\rho \approx 0$, not near the fitted $\hat{\rho}$ (vertical line). The likelihood’s preference for a large negative ρ reflects non-normality, not a bias the correction should apply.

over-adjusts; a semiparametric selection correction might behave better and is the natural next step. Second, sparsity: validated verbalization is rare (2.7%–24.8%), so the selected sample that identifies the outcome equation is small and sometimes empty (Qwen), which both inflates variance and makes the fit fragile.

The exclusion restriction is contestable despite randomization and the balance test: an instruction that changes what the model reports could in principle change the computation in ways our reliance measures miss — indeed it demonstrably does for the total effect R^{TE} , which is why we identify on the direct effect R^{NDE} , report R^{TE} only under ρ -sensitivity, and also give the mechanically instrument-free R^{pre} . Identification also leans on bivariate normality of (u, ε) ; non-normal latent reliance distorts $\hat{\rho}$. Our reliance measures are letter-level log-odds contrasts, not full computational attribution, and R^{TE} conflates through-CoT and direct paths (which is why R^{NDE} is reported). The verbalization detector is lexical; we bound its error by judge agreement. Scale is limited to $\leq 8\text{B}$ open models and one API model; hint types are four of the six in Chen et al. [1]. Finally, this is a measurement-methodology paper: even had the correction worked, it would repair an *estimate* of reliance, not make CoTs faithful.

8 What the negative teaches

Three points survive the negative result and, we think, are more useful than the positive we set out to find. (i) **The selection framing is the right diagnosis even where the fix fails.** Casting verbalization counts as a selected-sample statistic is correct: the ρ -test does detect selection, and the framework cleanly separates *is there selection* (the test, which works) from *how large is the corrected quantity* (the point estimate, which does not). Faithfulness audits should carry the test as a warning light even if they do not trust the correction. (ii) **The binding problem is measuring verbalization, not correcting for it.** A lexical detector overcounts verbalization by 2.2–9.5 $\times$ against a reading judge; since faithfulness rates are ratios built on V , this is a first-order error that no downstream correction addresses, and it recommends judge-based or otherwise validated V for any faithfulness measurement [12]. (iii) **Parametric selection correction is the wrong tool for heavy-tailed reliance.** The Gaussian IMR overshoots because log-odds reliance is non-normal; a semiparametric or bounds-based correction is the natural next step, and until one is validated, ρ -sensitivity bounds are the honest deliverable.

9 Reproducibility

All code, committed raw run records (JSONL), results JSONs, and this paper regenerate end-to-end: `numbers.tex` is machine-generated (`paper/gen_paper_numbers.py`; byte-identity checked by `paper/verify_regen.py`);

figures regenerate from committed raw data; the estimator core is gated by 36 tests including agreement with `statsmodels` and synthetic parameter recovery. Total compute: single-digit RTX-3080 GPU-hours plus one API model.

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